MSG No. 122.334 $I\bar{4}2d1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-4..1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,6,17,20,21,22]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,3,7,8,18,19,23,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,12,13,14,25,28,29,30]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[10,11,15,16,26,27,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,4,5,6,17,20,21,22]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,7,8,18,19,23,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,12,13,14,25,28,29,30]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[10,11,15,16,26,27,31,32]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,17,20]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}-z\right]$	`[2,3,18,19]`
3	$[0,\ 0,\ -z]$	`[5,6,21,22]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{4}\right]$	`[7,8,23,24]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,25,28]`
6	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}-z\right]$	`[10,11,26,27]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,14,29,30]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{3}{4}\right]$	`[15,16,31,32]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1,2,17,18]`
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[3,4,19,20]`
3	$\left[\frac{1}{4},\ -x,\ \frac{7}{8}\right]$	`[5,15,21,31]`
4	$\left[\frac{3}{4},\ x,\ \frac{7}{8}\right]$	`[6,16,22,32]`
5	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ \frac{3}{8}\right]$	`[7,13,23,29]`
6	$\left[\frac{1}{4},\ x+\frac{1}{2},\ \frac{3}{8}\right]$	`[8,14,24,30]`
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{5}{8}\right]$	`[9,10,25,26]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[11,12,27,28]`

Table 5: Wyckoff site: `16e`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{4}-z\right]$	`[2,18]`
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{4}-z\right]$	`[3,19]`
4	$[-x,\ -y,\ z]$	`[4,20]`
5	$[y,\ -x,\ -z]$	`[5,21]`
6	$[-y,\ x,\ -z]$	`[6,22]`
7	$\left[-y,\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	`[7,23]`
8	$\left[y,\ x+\frac{1}{2},\ z+\frac{1}{4}\right]$	`[8,24]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,25]`
10	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}-z\right]$	`[10,26]`
11	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}-z\right]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[13,29]`
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,30]`
15	$\left[\frac{1}{2}-y,\ -x,\ z+\frac{3}{4}\right]$	`[15,31]`
16	$\left[y+\frac{1}{2},\ x,\ z+\frac{3}{4}\right]$	`[16,32]`